

### KU5DSECSE309 CLOUD, EDGE AND FOG COMPUTING

| Semester | Course Type | Course Level | Course Code  | Credits | Total Hours |
|----------|-------------|--------------|--------------|---------|-------------|
| 5        | DSE         | 300          | KU5DSECSE309 | 4       | 60          |

| Learning Approach (Hours/ Week) |                       |          | Marks Distribution |     |       | Duration of ESE (Hours) |
|---------------------------------|-----------------------|----------|--------------------|-----|-------|-------------------------|
| Lecture                         | Practical/ Internship | Tutorial | CE                 | ESE | Total |                         |
| 4                               | 0                     | 1        | 50                 | 50  | 100   | 2                       |

#### Course Description:

"Cloud, Edge, and Fog Computing" is a comprehensive course aimed at exploring the transformative potential of distributed computing paradigms in the digital age. The course delves into the core concepts of Cloud Computing, elucidating its role as the backbone of modern IT infrastructure. The course introduces Edge Computing as a pivotal paradigm for reducing latency and enhancing responsiveness in distributed systems. The course provides the awareness about Fog Computing as an intermediate layer between the cloud and edge, facilitating decentralized data processing and analytics.

**Course Objectives:**

- Explore the need of new computing paradigms.
- To introduce concepts of Cloud, Edge and Fog Computing.

**Course Outcomes:**

At the end of the Course, the Student will be able to:

| SL # | Course Outcomes                                       |
|------|-------------------------------------------------------|
| CO1  | Obtain the knowledge about Cloud Computing.           |
| CO2  | To acquire knowledge on the concept of virtualization |
| CO3  | Obtain the knowledge about Edge computing             |
| CO4  | To understand the concept of Fog Computing            |

**Mapping of COs to PSOs**

|     | PSO1 | PSO2 | PSO3 | PSO4 | PSO5 |
|-----|------|------|------|------|------|
| CO1 | ✓    |      | ✓    |      | ✓    |
| CO2 | ✓    | ✓    |      | ✓    |      |
| CO3 | ✓    |      | ✓    |      | ✓    |
| CO4 | ✓    |      | ✓    |      | ✓    |

## **COURSE CONTENTS**

**Module 1:** Introduction to Cloud Computing, Recent Trends in Computing Cloud Computing, Evolution of cloud computing. Cloud deployment models: public, private, hybrid, community – Categories of cloud computing: Everything as a service: Infrastructure, platform, software, Security as a Service – Pros and Cons of cloud computing – Implementation levels of virtualization – virtualization structure – virtualization of CPU, Memory and I/O devices, Desktop Virtualization – virtual clusters and Resource Management – Virtualization for data center automation. ( 20 hours)

**Module 2:** Cloud Computing Architecture, Service Management in Cloud Computing Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), Data Management in Cloud Computing, Resource Management in Cloud Computing, Cloud Implementation. (20 hours)

**Module 3:** Edge computing: Introduction - Relevant Technologies - Fog and Edge Computing Completing the Cloud - Hierarchy of Fog and Edge Computing - Business Models – Edge Computing Platforms -Opportunities and Challenges. (20 hours)

**Module 4:** Challenges in Federating Edge Resources, Case for Optimization in Fog Computing-Formal Modeling Framework for Fog Computing – Metrics - Further Quality Attributes - Optimization Opportunities along the Fog Architecture - Optimization Opportunities along the Service Life Cycle. (20 hours)

**Module X:** Case study on the Tools and Products available for Virtualization. Taxonomy of Optimization Problems in Fog Computing. (20 hours)

## **Core Compulsory Readings**

1. Kai Hwang, Geoffery C. Fox and Jack J. Dongarra, "Distributed and Cloud Computing: Clusters, Grids, Clouds and the Future of Internet", First Edition, Morgan Kaufman Publisher, an Imprint of Elsevier, 2012.
2. Cloud Computing: Principles and Paradigms, Editors: Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Wiley, 2011.
3. Taheri J. & Deng S. (eds.): "Edge Computing: Models, technologies and applications", IET, 2020

## **Core Suggested Readings**

1. Gautam Shroff, Enterprise Cloud Computing - Technology, Architecture, Applications, Cambridge University Press, 2010.
2. Jason Venner, "Pro Hadoop- Build Scalable, Distributed Applications in the Cloud", A Press, 2009
3. Al-Turjman F. (ed.): "Edge Computing: from hype to reality", Springer, 2019
4. Kris Jamsa, Cloud Computing: SaaS, PaaS, IaaS, Virtualization, Business Models, Mobile, Security and more, Jones & Bartlett Learning Company, 2013
5. R. Buyya, C. Vecchiola, S T. Selvi, Mastering Cloud Computing, McGraw Hill (India) Pvt Ltd., 2013

## **TEACHING LEARNING STRATEGIES**

- Lecturing, Visualization, Team Learning, Lab session

## **MODE OF TRANSACTION**

- Lecture and Lab , Seminar, Discussion, Questioning and Answering

## ASSESSMENT RUBRICS

Refer to section 7 of FYIMP- Computational Science - Scheme and Syllabus for the 4 credit courses ( 4 Credit theory).

### **Sample Questions to test Outcomes.**

1. What are the characteristics of Cloud Computing?
2. What is Cloud Storage?
3. What is Data Virtualization?
4. Write a short paragraph on Cloud Design Principles.
5. Write a paragraph on emerging trends in Cloud Computing.
6. Cloud computing service models arranged as layers in a stack. Explain the meaning of the statement.
7. What are the limitations of Cloud Computing?
8. What do you mean by Online File storage?

## KU5DSECSE310 GEOGRAPHICAL INFORMATION SYSTEM

| Semester | Course Type | Course Level | Course Code  | Credits | Total Hours |
|----------|-------------|--------------|--------------|---------|-------------|
| 5        | DSE         | 300          | KU5DSECSE310 | 4       | 75          |

| Learning Approach (Hours/ Week) |                       |          | Marks Distribution |     |       | Duration of ESE (Hours) |
|---------------------------------|-----------------------|----------|--------------------|-----|-------|-------------------------|
| Lecture                         | Practical/ Internship | Tutorial | CE                 | ESE | Total |                         |